

Control Design for Morphing Aircraft Based on Frequency-Apportioned Control Allocation

Hanna Lee, Jihoon Lee, Seong-hun Kim, Seungyun Jung, Youdan Kim¹

Seoul National University, Seoul, Republic of Korea

Abstract. A control design strategy is proposed based on control allocation for the longitudinal motion of aircraft with variable-span and variable-sweep morphing wing. A morphing aircraft can change its shape during the flight, which allows the morphing aircraft to carry out various missions efficiently. Morphing parameters can be considered as control inputs, and therefore the system has redundant control inputs. The dynamics of morphing aircraft is transformed into a state-dependent coefficient form and regarded as an over-actuated system by augmenting control input with morphing parameters. Control allocation is used to distribute the generated control input adequately. Control allocation solves the optimization problem to minimize the displacement of control input, which is designed separately with the controller. Numerical simulations are performed to demonstrate the effectiveness of the proposed controller.

Keywords: variable-span, variable-sweep, morphing aircraft, unmanned aerial vehicle (UAV), flight control, frequency-apportioned control allocation (FACA)

1. Introduction

A morphing aircraft is an aircraft capable of large-scale shape changes [1]. The morphing system can be used to adapt to the varying flight phases or flight conditions, which may improve the flight performance such as mission efficiency, controllability or maneuverability. Recently, because of the versatilities, lots of research on morphing have been conducted, and various geometric approaches on morphing have been studied, which include the morphing wings of asymmetric span [2], variable camber [3], and gull wing [4]. In [5], open-loop control law to control the morphing parameters was proposed and the aerodynamic performance was analyzed. However, the open-loop control scheme could not guarantee the stability of the system. Various robust and adaptive control schemes were also studied for morphing aircraft control [6].

In this study, span and sweep angle variations are considered as morphing parameters, and controller is designed. A closed-loop control law is proposed where morphing parameters are considered as additional control effectors in addition to the conventional control surfaces. Therefore, the aircraft system becomes an over-actuated system which has redundant control effectors which can be used to generate the desired forces and moments. Control allocation (CA) method is utilized to distribute the desired forces and moments to the control effectors [7]. Various approaches of control allocation have been developed [8], for example, weighted pseudo-inverse method, daisy chain control allocation, direct control allocation, dynamic control allocation, and linear programming [9]. In this study, the pseudo-inverse method is adopted to solve the CA problem, which provides exact solution of

¹ Y. Kim (✉)

Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea
e-mail: ydkim@snu.ac.kr

optimization problem minimizing displacements of the control inputs. Note that the moments are generally considered as the subject to be distributed to the control effectors, but in this study, forces are also considered to exploit the property of morphing which can generate additional forces.

The frequency characteristics of each control effector should be considered in the CA, because the rates of morphing parameters are significantly slower than that of the conventional flapped control surfaces. In this study, the frequency-apportioned control allocation (FACA) proposed by Davidson et al. is adopted to account for the bandwidth gap between the conventional and morphing actuators [10]. The morphing aircraft dynamics changes according to the variation of morphing parameters and therefore, the system is described as a linear input-varying system that has a dependency on the variation of morphing parameters [11].

In summary, the control law considering system variation for the variable-span and variable-sweep morphing wing unmanned aerial vehicle (UAV) is proposed based on the FACA method. To demonstrate the effectiveness of the proposed scheme, numerical simulation is performed.

This paper is organized as follows. Longitudinal dynamic model is described in Section 2, and section 3 presents the control law design and control allocation method. Numerical simulation and discussion are described in Section 4. Finally, the conclusion is presented in Section 5.

2. Problem Statement

The morphing wing UAV considered in this study is shown in Fig. 1. Variable-span and variable-sweep morphing can be parametrized by two parameters; η_1 and η_2 . Span and sweep angle variations are linearly mapped onto $[-0.5, 0.5]$, which are stated in Table 1.

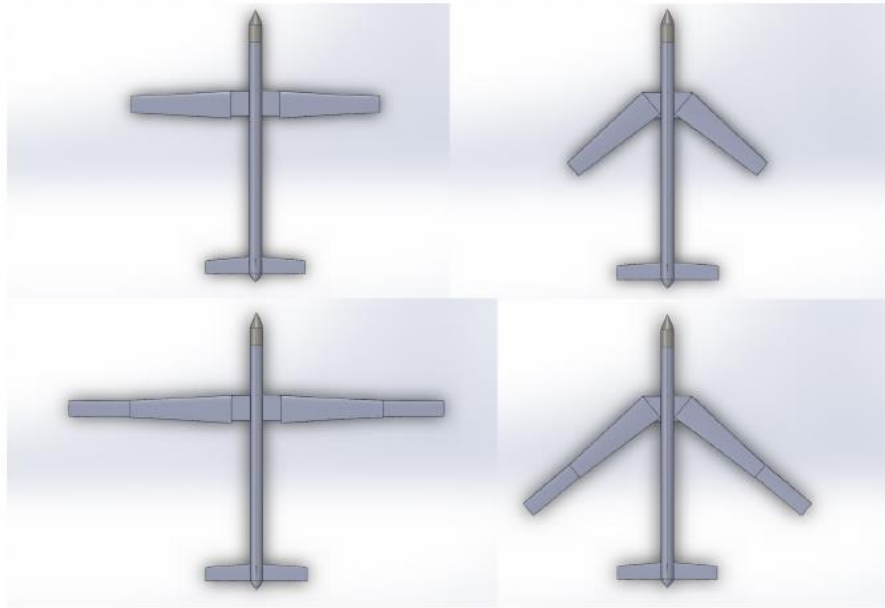


Fig. 1 Morphing wing UAV

Table 1 Morphing wing variable

	Min. Span	Max. Span	Min. Sweep	Max. Sweep
Value	1.7m	2.8m	0°	40°
Variable	$\eta_1 = -0.5$	$\eta_1 = 0.5$	$\eta_2 = -0.5$	$\eta_2 = 0.5$

2.1 Longitudinal Dynamic Model of Aircraft

In this study, only the longitudinal motion of aircraft is considered. The nominal state-space model is obtained at the flight condition of airspeed 20 m/s, with the altitude 300 m, where both morphing parameters are zero. The dynamics of the longitudinal model can be represented as follows,

$$\dot{x} = Ax + Bv \quad (1)$$

where $x = [V_T \quad \alpha \quad q \quad \gamma]^T$, $v = [F_x \quad F_z \quad M]^T$, and

$$A = \begin{bmatrix} -0.1057 & 6.9332 & 0 & -9.8067 \\ -0.0489 & -6.6621 & 1.0031 & 0 \\ 0 & -44.1712 & 0 & 0 \\ 0 & 0 & 1.0000 & 0 \end{bmatrix}, B = \begin{bmatrix} 0.1000 & 0 & 0 \\ 0 & 0.0050 & 0 \\ 0 & 0 & 0.1070 \\ 0 & 0 & 0 \end{bmatrix} \quad (2)$$

State vector x includes x-axis speed V_T , angle of attack α , pitch angle rate q , and flight path angle γ , and the input vector v is composed of x-axis force F_x , z-axis force F_z , and pitching moment M . [12]

2.2 Linear Input-Varying Model

The nominal dynamic model in section 2.1 consists of constant matrices, where the variation of dynamics according to morphing is not considered. To contain the effect of the variation, the nominal dynamic model is modified as follows,

$$\begin{aligned} \dot{x} &= A(\eta)x + Bv \\ &= A_0x + B_1(x)v \end{aligned} \quad (3)$$

where

$$A_0 = A|_{\eta=0} \quad (4)$$

$$B_1(x) = \left[\frac{\partial A}{\partial \eta_1} H_1 x + \frac{\partial A}{\partial \eta_2} H_2 x + B \right] \quad (5)$$

where

$$\eta_1 = H_1 v, \quad \eta_2 = H_2 v \quad (6)$$

Note that the partial derivatives of system matrix A depending on the morphing parameters are introduced. The system matrix A and B are modified to include the effect of morphing parameter variation as shown in the Eqs (4) and (5).

As a result, the linear system is transformed into a nonlinear system, which is called an input-varying system. The dynamic model of the system is formulated as a state-dependent system with state-dependent coefficient matrices.

Especially, input variables u are thrust, elevator angle, span parameter, and sweep parameter. However, in this part, virtual input variables are used. Note that v denotes the virtual input that represents x-axis force, z-axis force, and pitching moment, which are desired values for maneuvering of the morphing aircraft. The desired values are obtained by control law, and then they are distributed to input variables through CA.

3. Control Design

In this section, controller using control allocation method is designed for the state-dependent system. State-dependent Riccati equation (SDRE) strategy, a well-known optimal control scheme, is used [13, 14], and FACA is applied to the controller. FACA is one of the CA methods, which distributes the control forces and moments to control effectors considering the difference of frequency property.

3.1 Virtual Control Law

In this study, the control system is divided into controller and control allocator, and the SDRE is used for the virtual control law. The SDRE methodology uses the linear quadratic regulator (LQR) to formulate the nonlinear optimal control problem. The virtual control input is obtained by solving the algebraic Riccati equation (ARE), which is an extended linearization control method to deal with a nonlinear regulation problem. The considered performance index is defined as follows,

$$\min J = \int_0^{\infty} (x^T Q x + v^T R v) dt \quad (7)$$

subject to

$$\dot{x} = Ax + Bv \quad (8)$$

where the matrices Q and R are weighting matrices that can be chosen as diagonal matrices by considering the dimensions of the state and control effector.

By solving the optimization problem to minimize the performance index the optimal control input of virtual control law is obtained as

$$v = -R^{-1}(x)B^T(x)P(x)x \quad (9)$$

where $p(x)$ is the positive-definite solution of following SDRE.

$$P(x)A(x) + A^T(x)P(x) - P(x)B(x)R^{-1}(x)B^T(x)P(x) + Q(x) = 0 \quad (10)$$

Thus, the system dynamics becomes

$$\dot{x} = [A(x) - B(x)R^{-1}(x)B^T(x)P(x)]x \quad (11)$$

3.2 Control Allocation

Proposed input-varying system considering morphing parameters as the control effector is an over-actuated system, which has redundant control effectors. Thus, CA is required to distribute the obtained control inputs to the control effectors. By using CA, the control effectors are determined to provide the desired forces and moments.

In this study, the morphing parameters are assumed to be changed slowly compared to the conventional control surfaces, which means that there exists a difference between the rate of variation of morphing parameters and conventional control surfaces. The difference between those rates should be considered to determine control effectors.

The CA problem is to find the control input vector u , such that

$$Bu = v_{des} \quad (12)$$

subject to

$$u_{\min} \leq u \leq u_{\max} \quad (13)$$

where $v_{des} = [F_x \ F_z \ M]^T$ is the desired forces and moments determined by virtual control law, and B is the control effectiveness matrix of the following form.

$$B = \begin{bmatrix} \frac{\partial F_x}{\partial u_1} & \frac{\partial F_x}{\partial u_2} & \frac{\partial F_x}{\partial u_3} & \frac{\partial F_x}{\partial u_4} \\ \frac{\partial F_z}{\partial u_1} & \frac{\partial F_z}{\partial u_2} & \frac{\partial F_z}{\partial u_3} & \frac{\partial F_z}{\partial u_4} \\ \frac{\partial M}{\partial u_1} & \frac{\partial M}{\partial u_2} & \frac{\partial M}{\partial u_3} & \frac{\partial M}{\partial u_4} \end{bmatrix}, B \in \mathbb{R}^{m \times n} \quad (14)$$

The number of columns of B is greater than the number of rows of B for the over-actuated system. In this study, the number of columns is three; two longitudinal forces and pitching moment. And, the number of rows is four; throttle, elevator angle, and two morphing parameters for span and sweep angle variations.

Pseudo Inverse

The weighted pseudo-inverse method is used for CA method that requires a pseudo-inversion of the non-square matrix B . The solution of weighted pseudo-inverse is the two-norm solution to the CA and can be obtained by solving the following optimization problem.

$$\min J = \frac{1}{2} (u - c)^T W (u - c) \quad (15)$$

subject to

$$Bu = v_{des} \quad (16)$$

The solution can be obtained as follows,

$$u = -c + B^\dagger [v_{des} + Bc] \quad (17)$$

where c is the offset to consider the saturation of control effectors and $B^\dagger$ is the weighted pseudo-inverse of B as

$$B^\dagger = W^{-1} B^T (B W^{-1} B^T)^{-1} \quad (18)$$

where W is a diagonal positive definite weighting matrix and it can be selected to incorporate the saturation of the control effectors.

Frequency-Apportioned Control Allocation

The actuator rate limits should be considered to guarantee the control performance of the slowly varying system. The FACA method distributes i) the high-frequency component of the commands to the control effectors with higher rate limits which would be conventional control surfaces, and ii) the low-frequency component of the commands to the control effectors with lower rate limits which would be morphing parameters. The distribution is implemented by using a low-pass filter (LPF) and the distributed desired forces and moment with low frequency and high frequency are as follows,

$$v_h = [1 - L(s)]v_{des} \quad (19)$$

$$v_l = L(s)v_{des} \quad (20)$$

where the LPF is represented as

$$L(s) = \frac{1}{\tau s + 1} \quad (21)$$

The time constant τ is the design parameter that determines the frequency at which the commands are separated to the u_l and the u_h . The time constant can be chosen considering the rate limits of the morphing actuators.

To obtain the control input vector, the weighted pseudo-inverse method is used, and the solution is represented as,

$$u_h = B_h^\dagger v_h = B_h^\dagger [1 - L(s)]v_{des} \quad (22)$$

$$u_l = B_l^\dagger v_l = B_l^\dagger L(s)v_{des} \quad (23)$$

where $B_h^\dagger$ and $B_l^\dagger$ are high and low frequency weighted pseudo-inverse, respectively, as

$$B_h^\dagger = W_h^{-1} B_h^T (B_h W_h^{-1} B_h^T)^{-1} \quad (24)$$

$$B_l^\dagger = W_l^{-1} B_l^T (B_l W_l^{-1} B_l^T)^{-1} \quad (25)$$

where W_h and W_l are diagonal positive definite weighting matrices.

Consequently, the control input vector can be obtained as

$$u = u_l + u_h = [B_l^\dagger L(s) + B_h^\dagger \{1 - L(s)\}]v_{des} \quad (26)$$

where the control input vector $u = [\delta_t \ \delta_e \ \eta_1 \ \eta_2]^T$ is throttle, elevator, and morphing parameters.

A block diagram of the control allocator including algorithm of FACA is shown in Fig. 2, and a block diagram of the entire control system is shown in Fig. 3.

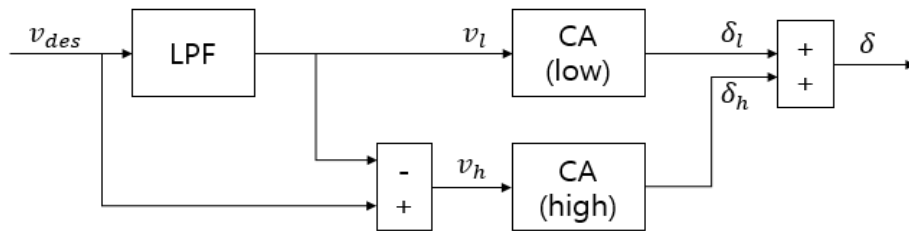


Fig. 2 Frequency-Apportioned Control Allocator

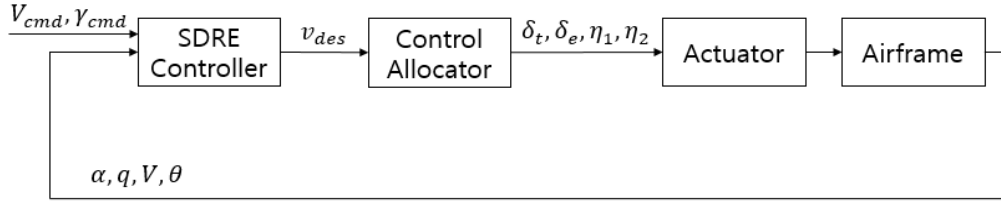


Fig. 3 Control System

4. Numerical Simulation

Numerical simulation is performed to demonstrate the performance of the proposed controller. The actuator dynamics are modeled considering the saturation of the actuator in the simulation, and the simulation time is set to 50 seconds. Simulation scenario 1 is summarized in Table 2, where the airspeed command is constant. Simulation scenario 2 is summarized in Table 3, where the flight path angle command is constant.

Table 2 Simulation Scenario 1

	0-10 sec	10-20 sec	20-30 sec	30-40 sec	40-50 sec
γ_{cmd}	0°	3°	0°	-3°	0°
V_{Tcmd}	20 m/s	20 m/s	20 m/s	20 m/s	20 m/s

Table 3 Simulation Scenario 2

	0-10 sec	10-20 sec	20-30 sec	30-40 sec	40-50 sec
γ_{cmd}	0°	0°	0°	0°	0°
V_{Tcmd}	20 m/s	18 m/s	20 m/s	25 m/s	20 m/s

First, the system that morphing parameters are included as control effectors is compared with the system that the morphing parameters are not included. Figure 4 shows the result of the scenario 1, which shows that when the morphing parameters are used as the control effectors, the use of the elevator is less than that without morphing. Instead of using the elevator, the span parameter is used as control effector. Also, the use of throttle is significantly reduced. Figure 5 shows the results of the comparison in scenario 2. It is shown that the use of throttle is saturated when the morphing parameters are not used. However, when the morphing parameters are used as the control effectors, the used of the throttle is maintained near the trim value and the deflection of the elevator is reduced. As shown in Fig. 4 and Fig. 5, the use of the conventional inputs is significantly reduced at the expense of slower convergence rate.

Second, the modified system considering the variation of dynamics by morphing is compared with the system without considering the morphing variation. The modified and nominal systems are controlled by LQR and SDRE based controller, respectively. Scenario 1 and scenario 2 are considered to compare the performance of the LQR and SDRE based controller. Figures 6 and 7 show that span and sweep parameters are more actively used for the flight path angle and the airspeed command tracking, respectively. It is also observed from Fig. 7 that the SDRE based controller shows faster convergence rate than LQR, which is partly because the former can include morphing-induced variations of the dynamics. According to the increase of altitude, span parameter is actively used. According to the increase of velocity, sweep parameter is actively used. Moreover, in both of the scenarios, the modified system using SDRE based controller provides more accurate fidelity than the existing system. Both of the performance indices of the scenarios are represented in Table 4. The performance index of the SDRE based controller is less than that of the LQR based controller.

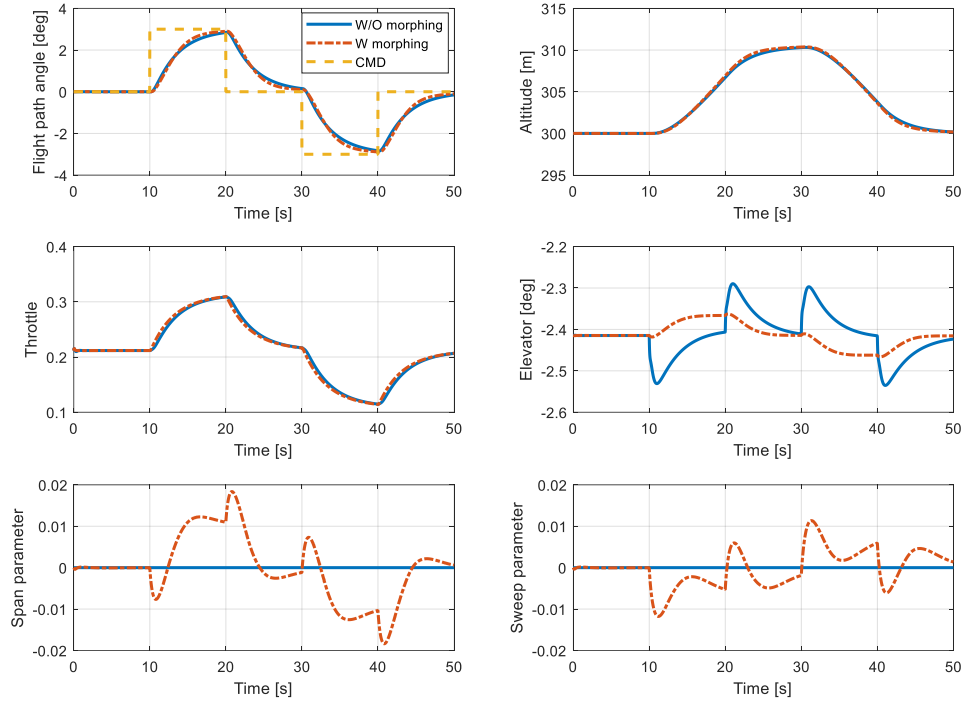


Fig. 4 Scenario 1: Flight path angle tracking

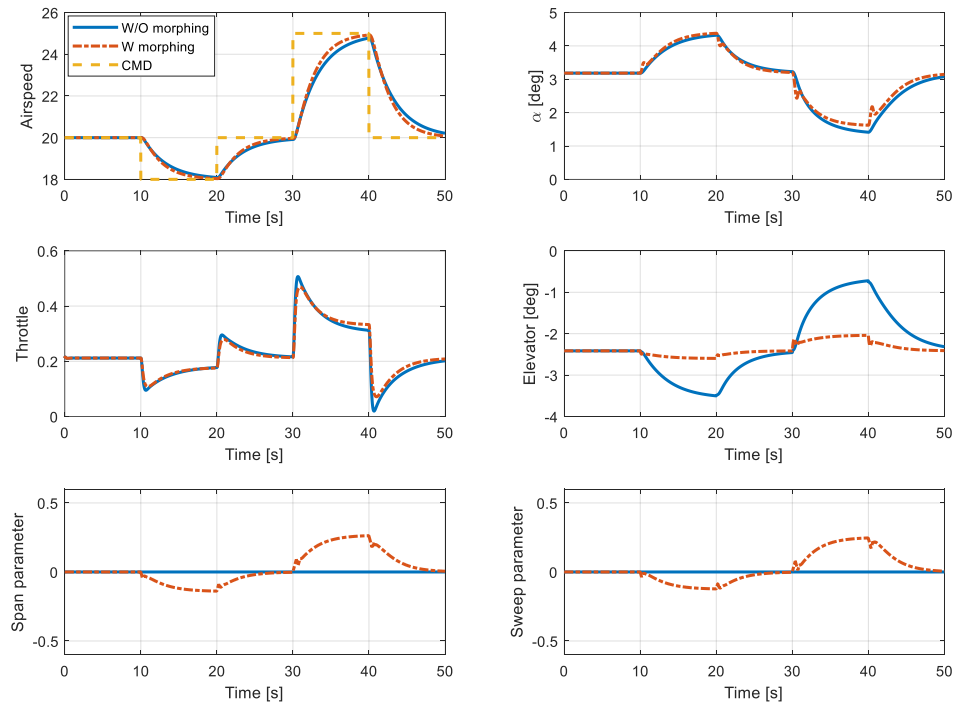


Fig. 5 Scenario 2: Airspeed tracking

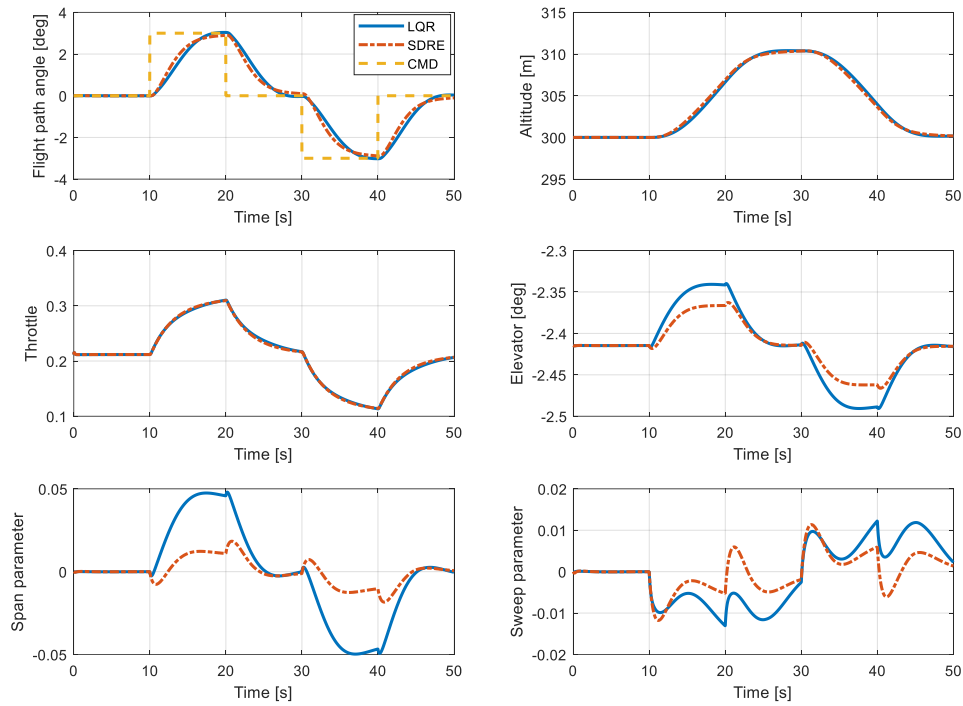


Fig. 6 LQR and SDRE controller for scenario 1

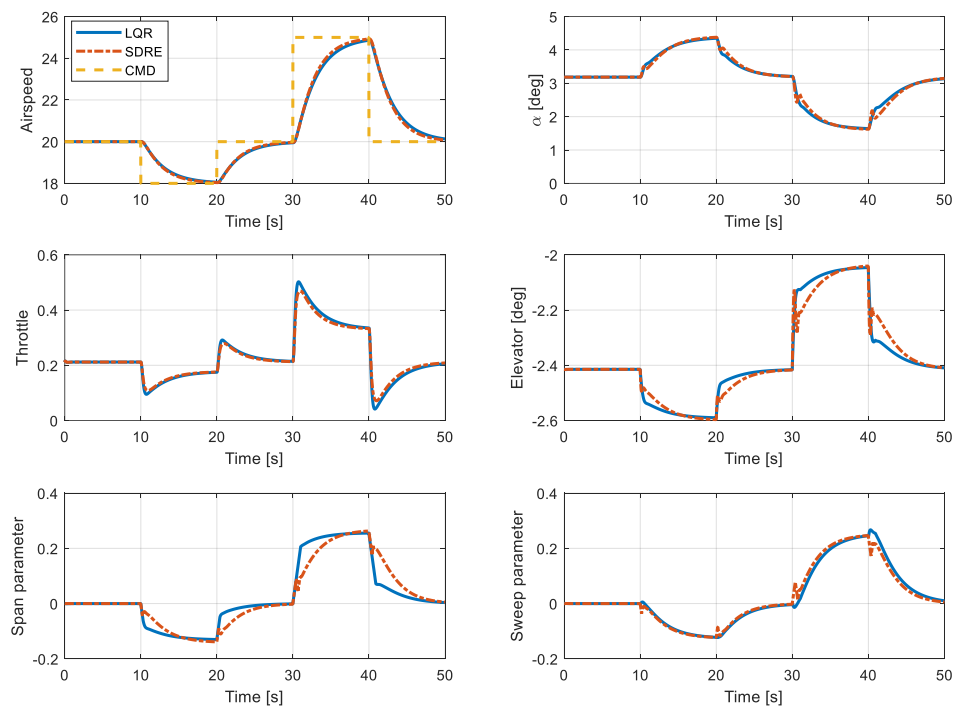


Fig. 7 LQR and SDRE controller for scenario 2

Table 4 Performance Index Comparison

	Scenario 1	Scenario 2
LQR	2.5197	4.4216
SDRE	2.4839	4.3009

To demonstrate of the performance of the FACA, simulation scenario 3 is considered and summarized in Table 5, where the airspeed command is constant and the flight path angle command is more aggressive than scenario 1. According to the cutoff frequency, the control effect is distributed low-frequency component and high-frequency component to each control allocator. As shown in Fig. 8, the elevator, which is a conventional control surface, has the high-frequency component, and the morphing parameters have the low-frequency component. As the cutoff frequency of LPF increase, the elevator moves fast.

In summary, because the variation of system dynamics by morphing is considered, the fidelity and operational performance are improved. Moreover, by considering the frequency difference of the conventional flap and morphing actuator, the rate limits of morphing parameters can be satisfied.

Table 5 Simulation Scenario 3

	0-10 sec	10-20 sec	20-30 sec	30-40 sec	40-50 sec
γ_{cmd}	0°	3°	-3°	3°	-3°
V_{Tcmd}	20 m/s	20 m/s	20 m/s	20 m/s	20 m/s

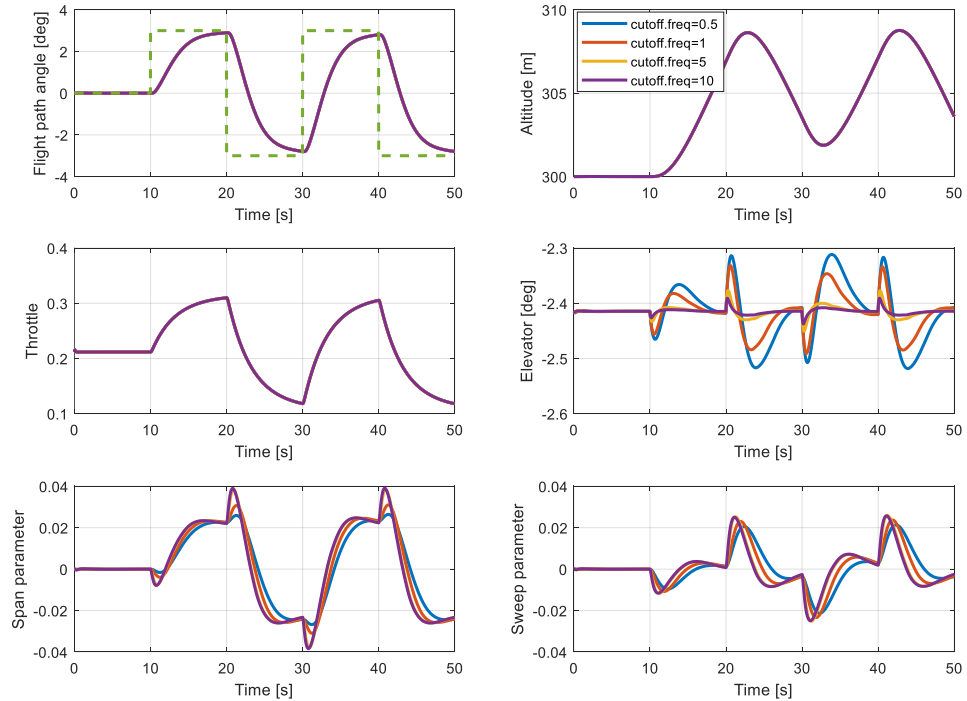


Fig. 8 Frequency-Apportioned Control Allocation for scenario 3

5. Conclusion

Control design strategy for a variable-span and variable-sweep morphing wing aircraft was proposed based on frequency-apportioned control allocation. The dynamic model of the morphing aircraft was transformed into an input-varying form. The required control efforts were distributed via the frequency-apportioned control allocation method to account for the low bandwidth of the morphing actuators. Numerical simulation was conducted to verify the effectiveness of the morphing aircraft. It was shown that morphing parameter produces additional forces and moments. For comparison, airspeed and flight path angle tracking controller using linear quadratic regulator and state-dependent Riccati equation-based controller was designed. Simulation results show that the morphing actuators can mitigate aggressive response and actuator saturation by complementing the conventional control surfaces, which makes the system more robust against uncertainties.

For the future studies, the lateral motion will be considered. Then, the couplings of forces and moments may extend flight envelope more. Moreover, this study could be applied to the fault-tolerant control scheme because of the increased degree of freedom of control effectors.

6. Acknowledgments

This work was supported by the R & D Program in Aerospace Parts and Materials (10066055) through the Korea Evaluation Institute of Industrial Technology funded by the Ministry of Trade, Industry, and Energy, Republic of Korea.

7. References

1. Ajaj RM, Beaverstock CS, Friswell MI. Morphing aircraft: The need for a new design philosophy. *Aerosp Sci Technol*. 2015;49: 154–166. Available from: <http://www.sciencedirect.com/science/article/pii/S1270963815003831>
2. Yue T, Zhang X, Wang L, Ai J. Flight dynamic modeling and control for a telescopic wing morphing aircraft via asymmetric wing morphing. *Aerosp Sci Technol*. 2017;70: 328–338. Available from: <http://dx.doi.org/10.1016/j.ast.2017.08.013>
3. Urnes JM, Morris C, Sheahan J, Dykman J, Clingman D. Control system design for a variable camber continuous trailing edge flap system on an elastic wing. In: *AIAA/ASME/ASCE/AHS/ASC Structures, structural dynamics, and materials Conf*. 2014. p. 1–10.
4. Obradovic B, Subbarao K. Modeling of flight dynamics of morphing gull-wing. *J Aircraft*. 2011;48(2): 391–402.
5. Prabhakar N, Prazenica RJ, Gudmundsson S, Balas MJ. Transient dynamic analysis and control of a morphing uav. In: *AIAA Guid Navig Control Conf*. 2016. p. 893–914.
6. Valasek J, Doebbler J, Tandale MD, Meade AJ. Improved adaptive-reinforcement learning control for morphing unmanned air vehicles. In: *IEEE Trans Syst Man, Cybernetics, Part B (Cybernetics)*. 2008. p. 1014–1020.
7. Durham WC. Constrained control allocation. *J Guid Control Dyn*. 1993;16(4): 717–725. Available from: <http://arc.aiaa.org/doi/10.2514/3.21072>
8. Oppenheimer MW, Doman DB, Bolender MA. Control allocation for over-actuated systems. *MED*. 2006;(2).
9. Johansen TA, Fossen TI. Control allocation - a survey. *Automatica*. 2013;49(5): 1087–1103. Available from: <http://dx.doi.org/10.1016/j.automatica.2013.01.035>
10. Lallman F, Davidson J, Bundick W. Integrated reconfigurable control allocation. In: *AIAA Guid Navig Control Conf Exhib*. 2001;
11. Boothe K, Fitzpatrick K, Lind R. Controllers for disturbance rejection for a linear input-varying class of morphing aircraft. In: *AIAA/ASME/ASCE/AHS/ASC Structures, structural dynamics, and materials conference*. 2005. p. 1–14.
12. Beard RW, McLain TW. *Small unmanned aircraft: theory and practice*. Princeton University Press; 2013.
13. Cimen T. State-dependent Riccati equation (SDRE) control: a survey. *IFAC Proceedings Volumes (IFAC-PapersOnline)*. IFAC. 2008;17: p. 3761–3775. Available from: <http://dx.doi.org/10.3182/20080706-5-KR 1001.00635>
14. Cloutier JR. State-dependent Riccati equation techniques: an overview. *Proc Am Control Conf*. 1997. p. 932–936.